
Customer Churn Patterns - SyriaTel

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Roadmap

1. Business Problem
2. Data Summary and Preparation
3. Modeling and Analysis
4. Summary and Recommendations



Business Understanding

- Company wants to minimize customer churn
- Customer Acquisition is more expensive than Customer Retention
 - Questions for us:
 - a. **Are there any patterns we can observe in customers churning?**
 - Can we do anything to intervene on these patterns?

Data Understanding

- Dataset
 - a. 3,333 Customers, 21 Columns
 - b. Removing unnecessary columns leaves 18 factors for us to consider
- **TARGET:** “Churn” column
- False Negatives more significant than False Positives
 - a. A positive prediction entails that the customer is likely to churn.
 - b. A negative prediction indicates a customer is not likely to churn.

Metrics: Recall Score

High Recall Score =

Low False Negative Rate

Data Preparation

- Scaling the Data
 - a. More accurate reading of impact of different factors
- Splitting the Data into different sets
 - a. Allows for multiple testing runs to minimize impact of “flukes”
- Model Experimentation
 - a. Testing different predictive models to see **which has the highest Recall score**

Data Modeling & Analysis

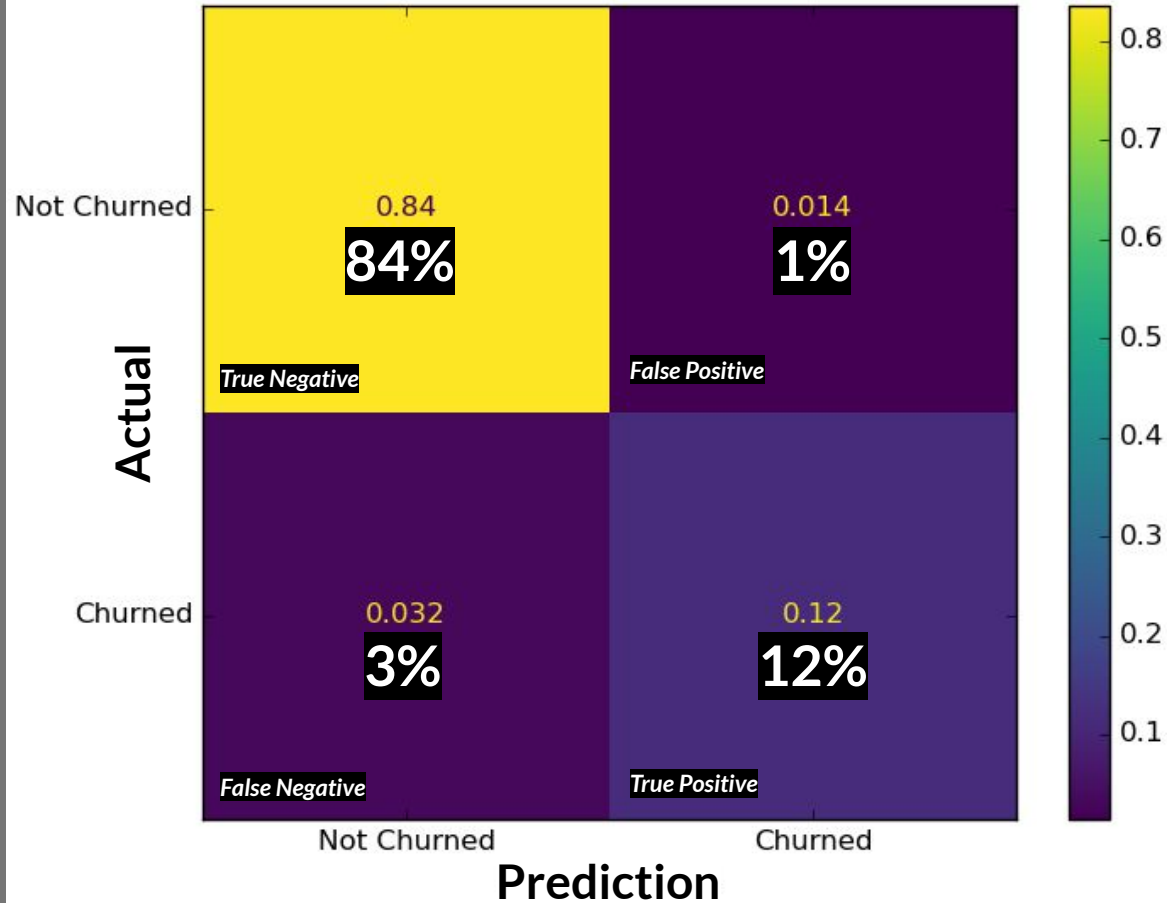
Modeling

Critical Features:

- Use for tricky relationships
- Useful for determining important of factors

Random Forest Classifier

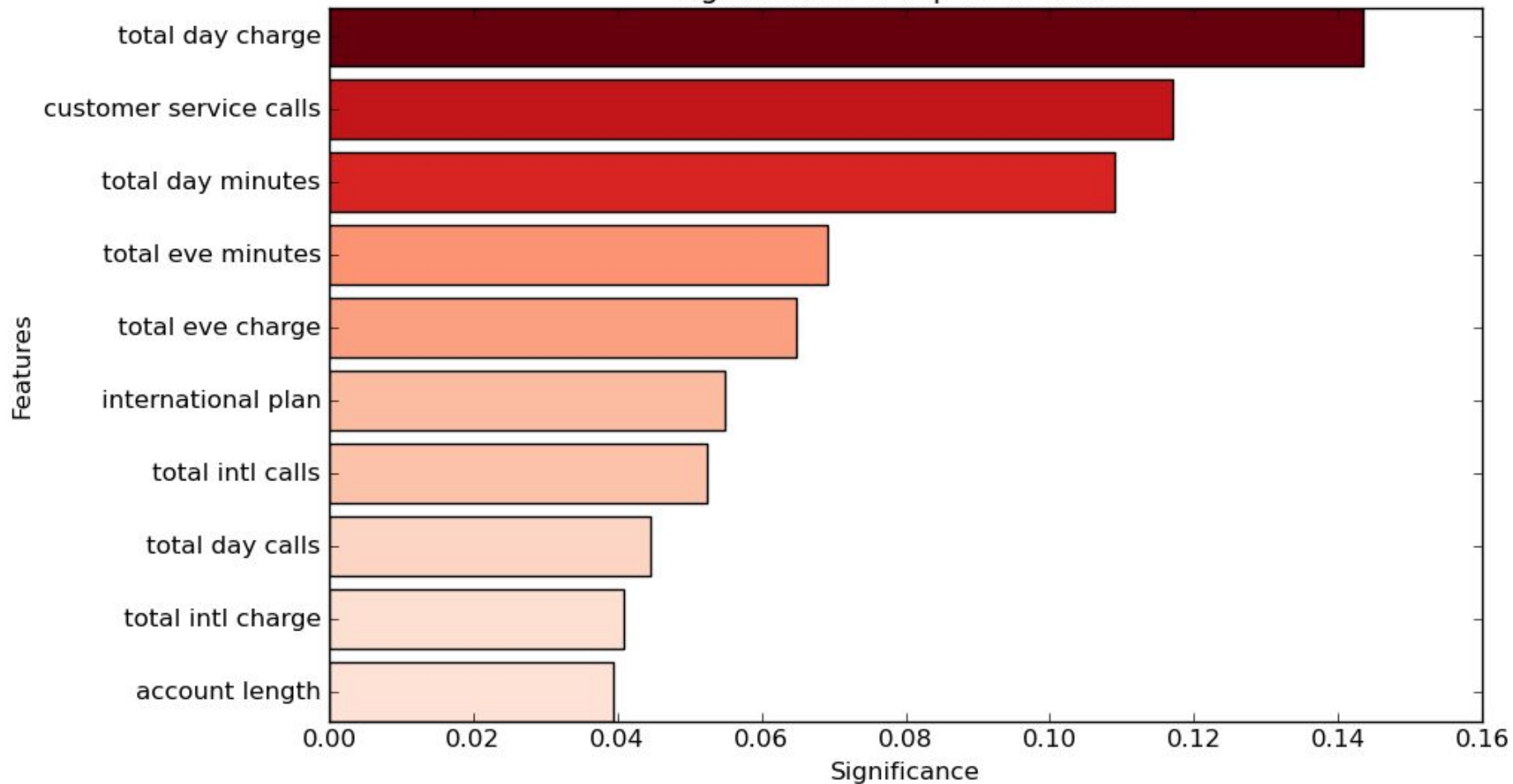
**How successful was
our model?**



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**Which factors were
most significant in
determining churn?**

Significance of Top 10 Features



Conclusion

Factor Significance

1. *Day Calls*: Time and Charges
2. *Customer Service Calls*: number of times customer service was contacted

Limitations:

- Timing of customer churn
 - a. Related to our policy changes?
 - b. Related to competitor policy changes?

Next Steps

**Revisiting
Customer
Service
Protocols**

**Reconsidering
Charges,
Accommodations,
and Plans for Day
Calls**

Future Investigation:

Looking at timing of
churns and potential
overlap with policy
changes

Questions?

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